

# Besher Alkurdi

NLP Research Engineer



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  Jyväskylä, Finland (Relocatable)

## Experience

- Research Engineer at Huawei** Aug 2023 - Aug 2024
- Enhanced custom in-house distributed training library with parameter-efficient methods, context expansion, and training stabilization techniques, achieving 8x resource reduction without performance degradation.
  - Conducted customer requirements analysis sessions, architected ML solutions for business needs, and defined PoC implementation roadmaps with cross-functional teams.
  - Coded LLM architectures enabling compatibility with standard libraries, delivering significant inference speedup.
  - Built scalable multi-backend inference platform with RESTful/streaming APIs, supporting scalable deployment and advanced sampling methods.
  - Wrote multilingual synthetic data generation toolkit and fine-tuning pipelines for customer LLM adaptations.
  - Ran ZeRO/FSDP/DeepSpeed multi-node training, tested models and supplied teams with improved data pipelines.
- Undergraduate Research Fellow at Yildiz Technical University** Jun 2022 - Mar 2023
- Curriculum learning research in *publicly funded project*, implementing cyclical and difficulty-based training strategies.
  - Developed data augmentation pipeline and experimented with auxiliary classifiers for early exit strategies.
  - Deployed distributed training systems using TensorFlow, PyTorch, and Horovod on lab computers.
  - Trained CLIP-style multimodal models with contrastive learning.
- Research Intern at New Mind AI** Jul 2021 - Oct 2021
- Conducted research on state-of-the-art natural language processing techniques and their application in a system assisting lawyers in lawsuits focusing on Document AI applications in the Turkish language.

## Education

- MSc. in Artificial Intelligence** Thesis Phase  
University of Jyväskylä
- BSc. in Computer Science** Jun 2023  
Yildiz Technical University  
2nd in class, GPA: 3.84/4

## Publications

- Scaling Up Paraphrase Generation Datasets with MT and Semantic Similarity Filtering** 2024
- An extension of our prior work. Experimented with data augmentation using paraphrasing.
- Semantic Similarity Based Filtering for Turkish Paraphrase Dataset Creation** Dec 2022
- A method for creating paraphrase datasets from parallel corpora along the largest paraphrase datasets in Turkish.
- Linguistic-based Data Augmentation Approach for Offensive Language Detection** 2022
- Augment classification datasets with a linguistic-based approach improving the detection of offensive language.

## Skills

**ML/DL:** PyTorch, TensorFlow, HuggingFace, OpenCV, NLP, TRAX, ZeRO, FSDP, DeepSpeed

**Programming:** Python, C, Rust, SQL, Java, Julia

**Tools:** Linux, Git/GitHub, ROS, Docker

## Languages

**Arabic:** Native • **English:** Advanced • **Turkish:** Advanced • **Finnish:** Beginner • **French:** Beginner

## Certificates

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| <b>Scratch to Scale: Large-Scale Training</b>   <i>Maven</i> <ul style="list-style-type: none"><li>• DDP, ZeRO, Pipeline &amp; Tensor Parallelism, Multi-Dimensional Parallelism for LLM scaleups.</li></ul> | Oct 2025 |
| <b>Deep Reinforcement Learning Course</b>   <i>HuggingFace</i> <ul style="list-style-type: none"><li>• Q-Learning, PPO.</li></ul>                                                                            | May 2024 |
| <b>Generative Adversarial Networks (GANs) Specialization</b>   <i>DeepLearning.AI</i> <ul style="list-style-type: none"><li>• GAN components, data augmentation, privacy.</li></ul>                          | Jul 2023 |
| <b>Natural Language Processing Specialization</b>   <i>DeepLearning.AI</i> <ul style="list-style-type: none"><li>• NLP, Transformers, Attention models, Machine Translation.</li></ul>                       | Jun 2023 |
| <b>Deep Learning Specialization</b>   <i>DeepLearning.AI</i> <ul style="list-style-type: none"><li>• Deep learning, including GANs and VAEs.</li></ul>                                                       | Feb 2023 |
| <b>Machine Learning Specialization</b>   <i>DeepLearning.AI</i> <ul style="list-style-type: none"><li>• ML fundamentals and practical applications.</li></ul>                                                | Aug 2022 |
| <b>Introduction to Computational Thinking and Data Science</b>   <i>edX</i> <ul style="list-style-type: none"><li>• Introductory course in data science and ML.</li></ul>                                    | May 2021 |

**Verification:** All certificates are verifiable at <https://www.linkedin.com/in/mrbesher>.

## Volunteer Experience

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| <b>Volunteer Researcher</b> at COSMOS AI Research Group<br>Conducting research on improving the performance of large language and vision-language models using unlabeled data and reinforcement learning without reliance on stronger teacher supervision. | Dec 2024 – Present          |
| <b>Tutor</b> at Schoolhouse.world<br>Tutoring students worldwide in subjects such as Math, Computer Science, and more.                                                                                                                                     | Feb 28, 2021 - May 20, 2021 |
| <b>Tutor</b> at Aqsa Youth Institution<br>Taught computer, networking, and programming basics to middle school and high school refugee students.                                                                                                           | Nov 30, 2020 - Aug 31, 2021 |
| <b>Translator</b> at Ask Me<br>Assisted tourists visiting Istanbul as part of Fatih Municipality's "Voluntary Tourism Delegates" program.                                                                                                                  | Jun 30, 2019 - Aug 31, 2019 |